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AI Lab Week 1

Water Jug Problem

CODE:

```
[ ]: from collections import deque

[ ]: A = int(input("Enter amount of litres in jug A: "))
    B = int(input("Enter amount of litres in jug B: "))
    goal = int(input("Enter the the target for jug A: "))
    path_exists = False

[ ]: visited = set([(0,0)])
    queue = deque([(0,0)])
    parent = {(0,0) : None}
    ans_state = []

    while queue:
        state = queue.popleft()

        if state[0] == goal:
            path_exists = True
            ans_state = state
            break

        fill_A = (A, state[1])
        fill_B = (state[0], B)
        empty_A = (0, state[1])
        empty_B = (state[0], 0)

        new_B = min(state[0], B - state[1])
        A_to_B = (state[0] - new_B, state[1] + new_B)

        new_A = min(state[1], A - state[0])
        B_to_A = (state[0] + new_A, state[1] - new_A)

        generated_states = [fill_A, fill_B, empty_A, empty_B, A_to_B, B_to_A]
```

```
for i in generated_states:
    if i not in visited:
        queue.append(i)
        visited.add(i)
        parent[i] = state
```

```
if ans_state:
    path = [ans_state]
    i = ans_state
    while parent[i]:
        path.append(parent[i])
        i = parent[i]
    path.reverse()
```

```
for i in path:
    print(i)
```

```
print("Goal Reached")
```

```
else:
    print("No solution")
```

I/O

Input - 1:

```
Enter amount of litres in jug A: 4
Enter amount of litres in jug B: 3
Enter the the target for jug A: 2
```

Output-1:

```
(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
Goal Reached
```

Input - 2:

```
Enter amount of litres in jug A: 5
Enter amount of litres in jug B: 3
Enter the the target for jug A: 4
```

Output - 2:

```
(0, 0)
(5, 0)
(2, 3)
(2, 0)
(0, 2)
(5, 2)
(4, 3)
Goal Reached
```

Input - 3:

```
Enter amount of litres in jug A: 2
Enter amount of litres in jug B: 6
Enter the the target for jug A: 5
```

Output - 3:

```
No solution
```

Input - 4:

```
Enter amount of litres in jug A: 8
Enter amount of litres in jug B: 5
Enter the the target for jug A: 4
```

Output - 4:

```
(0, 0)
(8, 0)
(3, 5)
(3, 0)
(0, 3)
(8, 3)
(6, 5)
(6, 0)
(1, 5)
(1, 0)
(0, 1)
(8, 1)
(4, 5)
Goal Reached
```

Input - 5:

```
Enter amount of litres in jug A: 7
Enter amount of litres in jug B: 5
Enter the the target for jug A: 6
```

Output - 5:

```
(0, 0)
(7, 0)
(2, 5)
(2, 0)
(0, 2)
(7, 2)
(4, 5)
(4, 0)
(0, 4)
(7, 4)
(6, 5)
Goal Reached
```